Adam Larson
Data Engineer

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GitHub Profile

LinkedIn Profile

EDUCATION

B.S.
Computer Science
San Diego State University
2025
San Diego, CA

Relevant courses

- Database Theory & Implementation
- Operating Systems
- Systems Programming
- Cloud Computing
- Machine Learning

SKILLS

- Python (ETL, async)
- SQL (CTEs, window functions)
- FastAPI / REST APIs
- MongoDB (Motor async)
- Redis (caching, real-time)
- GCP (Storage, Secrets, Tasks, KMS)
- Docker + Linux
- Observability & Reliability

CAREER OBJECTIVE

Data Engineer building reliable data pipelines and backend services that turn raw data into usable datasets. Interested in ingestion, transformation, and automation for analytics-ready systems.

PROJECTS

Langlift – Language-learning platform backend (Personal Project)

- Designed and implemented a production-style backend for a language-learning platform, delivering **82 REST API** endpoints across content, community feeds, study flows and user features.
- Implemented real-time updates via Server-Sent Events (SSE) with subscription management and streaming endpoints to support live notifications and feed updates.
- Built async services with a MongoDB data model optimized for flexible multilingual card types, efficient user-scoped queries and indexed lookup paths.
- Implemented background and scheduled workflows (job queues + cron-style tasks).
- Integrated PropelAuth, Redis, OneSignal, and GCP services (Storage, Secrets, Tasks, Scheduler, Key Manager) with centralized lifecycle management, health checks, and error handling.
- Built a ServiceManager to initialize and manage **17 external services** with lifecycle hooks, health checks, and consistent error handling across the codebase.
- Added centralized structured logging to capture operational context without exposing sensitive data.

Spaceship Titanic – Kaggle tabular ML pipeline

- Built a leakage-safe ML pipeline using GroupKFold cross-validation on passenger groups derived from **PassengerId** and a shared feature pipeline for train/test parity.
- Engineered features from cabin parsing, spending behavior and group consistency signals; implemented group-aware imputation with indicator flags.
- Trained and compared strong tabular baselines (CatBoost and HistGradientBoosting) and documented error modes and feature importance.
- Packaged the project as a reproducible repo (Poetry, feature pipeline scripts, artifacts) and generated feature importance plots for interpretability.
- Achieved 0.804 Kaggle public leaderboard score (rank 627 at time of submission).

Kanji Trainer for Raspberry Pi

- Built a responsive touch-friendly spaced repetition app (SM-2) with daily progress tracking and offline-first usage on Raspberry Pi hardware.
- Designed SQLite databases including **FTS5** search; optimized lookup paths for **bilingual search** across two languages.
- Built an ingestion/ETL workflow that formats raw word lists and KanjiVG XML/SVG assets into normalized SQLite tables and searchable indexes for fast in-app retrieval.
- Implemented a custom on-screen keyboard and improved UI responsiveness by optimizing input/render loops and query behavior for low-power constraints.